

Computer Fundamentals Notes

1. BIOS (Basic Input Output System)

- BIOS is a firmware stored on the motherboard.
- It starts the computer when power is turned on.
- It checks hardware devices through POST (Power-On Self-Test).
- It loads the operating system into memory.
- Access BIOS by pressing **F2, F10, F12, DEL, or ESC** during startup (depends on manufacturer).

2. First Screen of Windows (Desktop)

- The first screen displayed after Windows starts is called the **Desktop**.
- It contains:
 - Icons
 - Taskbar
 - Start Button
 - Wallpaper
 - Notification Area

Desktop Shortcut Keys

Action	Shortcut
Show Desktop	Windows + D
Open File Explorer	Windows + E
Open Run	Windows + R
Open Settings	Windows + I

3. ROM (Read Only Memory)

- ROM is permanent memory.
- Data remains stored even when power is off.

- It stores BIOS and startup instructions.

- Types:
 - PROM
 - EPROM
 - EEPROM

4. Taskbar

- Located at the bottom of the desktop.
- Contains:
 - Start Button
 - Search Box
 - Pinned Applications
 - Notification Area
 - Clock & Date

Taskbar Shortcuts

Action	Shortcut
Open Task Manager	Ctrl + Shift + Esc
Switch Apps	Alt + Tab
Open Start Menu	Windows Key

5. Title Bar

- Appears at the top of every window.
- Displays the application or document name.
- Contains:
 - Minimize Button
 - Maximize/Restore Button
 - Close Button

6. Minimize, Maximize & Close

Minimize

- Hides window to taskbar.
- Shortcut: **Windows + Down Arrow**

Maximize

- Enlarges window to full screen.
- Shortcut: **Windows + Up Arrow**

Close

- Closes current window.
 - Shortcut: **Alt + F4**
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7. Software

Software is a set of programs that instructs a computer what to do.

Types of Software

1. System Software
 2. Utility Software
 3. Application Software
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8. System Software

- Controls and manages computer hardware.
- Provides a platform for application software.

Examples

- Windows
 - Linux
 - macOS
 - Android
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9. Utility Software

- Maintains and optimizes computer performance.

Examples

- Antivirus
 - Disk Cleanup
 - Backup Software
 - WinRAR
 - Disk Defragmenter
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10. Application Software

- Designed for user-specific tasks.

Examples

- MS Word
 - MS Excel
 - PowerPoint
 - Chrome
 - Photoshop
 - VLC Media Player
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11. Show Desktop

Method 1

Press **Windows + D**

Method 2

Click the small area at the extreme right corner of the taskbar.

Method 3

Right Click Taskbar → Show Desktop

12. Important Windows Shortcut Keys

Function	Shortcut
Copy	Ctrl + C
Cut	Ctrl + X
Paste	Ctrl + V
Undo	Ctrl + Z

Function	Shortcut
Redo	Ctrl + Y
Save	Ctrl + S
Select All	Ctrl + A
Print	Ctrl + P
Find	Ctrl + F
New Folder	Ctrl + Shift + N
Rename	F2
Refresh	F5
Delete	Delete
Permanent Delete	Shift + Delete
Task Manager	Ctrl + Shift + Esc
Lock PC	Windows + L
File Explorer	Windows + E
Run Command	Windows + R
Settings	Windows + I
Search	Windows + S
Screenshot	Windows + Shift + S
Switch Window	Alt + Tab
Close Window	Alt + F4

13. Shutdown Computer

Manual Method

1. Click Start Button
2. Power Icon
3. Shutdown

Shortcut Method

1. Press **Alt + F4**
2. Select Shutdown
3. Press Enter

Command Method

1. Open Run (**Windows + R**)
2. Type:
shutdown /s /t 0
3. Press Enter

14. Force Shutdown

Method 1

Press and hold the **Power Button** for 5–10 seconds.

Method 2

Task Manager → End Task (for frozen applications).

⚠ Use only when the computer is not responding.

15. Restart Computer

Manual Method

1. Start Button
2. Power Icon
3. Restart

Shortcut Method

1. Press **Alt + F4**
2. Select Restart
3. Press Enter

Command Method

1. Open Run (**Windows + R**)
2. Type:
shutdown /r /t 0
3. Press Enter